

	IILM University, Greater Noida	
Subject Name: Semiconductor and Quantum Physics	Course Code:	26SOST101
TUTORIAL		

UNIT 1

1. The forbidden gap in pure silicon is 1.1eV. Compare the number of conduction electrons at temperatures 37°C and 27°C.

Answer: the number of conduction electrons increases by approximately 2.96 times.

2. Compute the concentration of intrinsic charge carriers in a germanium crystal at 300K. Given that $E_g = 0.72\text{eV}$ and $m_{e^*} = m_e$.

Answer: $2.27 \times 10^{19} \text{ m}^{-3}$

3. A metal has a Fermi energy of 5.5 eV. Calculate the Fermi temperature.

Answer: $6.38 \times 10^4 \text{ K}$

4. Calculate the occupation probability of an energy level 0.02 eV above the Fermi level at 300 K using the Fermi–Dirac distribution function.

Answer: 0.315

5. Determine the occupation probability of an energy level 0.10eV above the Fermi level at 800K.

Answer: 0.19

6. At 300K find the energy level where the probability of finding an electron is 90% ($F(E) = 0.9$) in a metal with a Fermi energy of 5.5 eV.

Answer: 5.443eV

7. A superconducting tin has a critical temperature of 3.7K in zero magnetic fields and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2K.

Answer: 0.02166 Tesla

8. The transition temperature for Pb is 7.2 K. However, at 5 K it loses the superconducting property when subjected to a magnetic field of 3.3×10^4 A/m. Find the maximum value of H which will allow the metal to retain its superconductivity at 0 K.

Answer: $H_0 = 6.37 \times 10^4$ A/m

9. At what temperature is $H_c(T) = 0.1H_c(0)$ for lead (Pb) having a critical temperature $T_c = 7.2$ K?

Answer: 6.83 K

10. The critical fields at 6 K and 8 K for NbTi alloys are 7.616 and 4.284 mA/m respectively. Calculate the transition temperature and critical field at 0 K.

Answer: 11.9 mA/m, $T_c = 10$ K

11. For a specimen of superconductor, the critical fields are 1.4×10^5 and 4.2×10^5 A/m for temperature 14 K and 13 K. Calculate the transition temperature and critical fields at 0 K and 4.2 K.

Answer: $T_c = 14.474$ K, $H_0 = 2.173 \times 10^6$ A/m, $H_c(4.2 \text{ K}) = 1.99 \times 10^6$ A/m

12. Determine the transition temperature and critical field at 4.2 K for a given specimen of superconductor if the critical fields are 1.410×10^5 and 4.205×10^5 amp/m at 14.1 K and 12.9 K respectively.

Answer: $T_c = 14.668$ K, $H_0 = 1.856 \times 10^6$ A/m, $H_c(4.2 \text{ K}) = 1.704 \times 10^6$ A/m

UNIT 2

1. An electron in a semiconductor has a crystal momentum $p = 3.0 \times 10^{-25} \text{ kg-m/s}$. Calculate its energy in Joules and in eV.

Answer: $E = 4.94 \times 10^{-20} \text{ J}$ (or 0.308 eV).

2. Find the propagation constant k of an electron in a crystal lattice if its kinetic energy is 1.5 eV.

Answer: $k = 6.272 \times 10^9 \text{ m}^{-1}$.

3. Calculate the energy (in Joules and in eV) of an electron in a semiconductor conduction band having a propagation constant $k = 2.0 \times 10^9 \text{ m}^{-1}$.

Answer: $E = 2.443 \times 10^{-20} \text{ J}$ (or 0.1525 eV).

4. Calculate the forward current flowing through a Germanium P-N junction diode (ideality factor $n = 1$) at temperature $T = 300 \text{ K}$ when a forward bias voltage of 0.2 V is applied. The reverse saturation current of the diode is $2.0 \mu\text{A}$.

Answer: Current $I = 4.591 \text{ mA}$.

5. A Silicon P-N junction diode (ideality factor $n = 2$) operates at 300 K with a reverse saturation current $I_R = 10 \text{ nA}$. Find the total current through the diode when a forward voltage $V = 0.6 \text{ V}$ is applied.

Answer: Current $I = 1.1005 \text{ mA}$.

6. Determine the reverse current flowing through a Silicon diode ($n = 2$) when a reverse voltage $V = -0.5 \text{ V}$ is applied at 300 K. The reverse saturation current is $I_R = 5.0 \text{ nA}$.

Answer: Reverse Current $I \approx -5.0 \text{ nA}$ (equal in magnitude to $-I_R$).

7. A photovoltaic cell receives a total incident solar power $P_{\text{in}} = 10.0 \text{ W}$. If the short circuit current $I_{\text{sc}} = 3.0 \text{ A}$, open circuit voltage $V_{\text{oc}} = 0.6 \text{ V}$, and fill factor $FF = 0.80$, calculate the efficiency (η) of the cell.

Answer: Efficiency $\eta = 14.40\%$.

8. A photovoltaic cell with an efficiency of 14% is illuminated by solar radiation supplying an incident power $P_{\text{in}} = 10.0 \text{ W}$. Given that the short circuit current $I_{\text{sc}} = 2.5 \text{ A}$ and fill factor $FF = 0.75$, calculate the open circuit voltage V_{oc} .

Answer: Open Circuit Voltage $V_{oc} = 0.7467 \text{ V}$.

9. A solar cell has a short circuit current $I_{sc} = 4.0 \text{ A}$, open circuit voltage $V_{oc} = 0.65 \text{ V}$, and fill factor $FF = 0.75$. If the efficiency of the solar cell is 13.0%, determine the total incident solar power P_{in} required.

Answer: Required Incident Solar Power P , $P_{in} = 15.0 \text{ W}$.

10. Find the efficiency of a single solar cell (10 cm x 10 cm) which produces a voltage of 0.5 V and a current up to 2.5 A. Given, the solar insolation is 800 Wm^{-2} .

Answer: $\eta = 15.625\%$ (or 15.63%).

11. A single solar cell on illumination by insolation of about 800 Wm^{-2} produces a voltage of 0.5 V and a current up to 2.0 A. Calculate the area of the cell, given the efficiency of the solar cell is 12.5%.

Answer: Area = 0.01 m^2 (or 100 cm^2).

12. The solar insolation on a rectangular module (1.5 m x 2.0 m) of a photovoltaic cell is 550 Wm^{-2} . Calculate the power output of the module if the efficiency of the cell is 12%.

Answer: Power Output = 198 W.

UNIT 3

1. Calculate the de-Broglie wavelength associated with a proton moving with a velocity equal to $1/20^{\text{th}}$ of the velocity of light. Given the mass of the proton is $1.67 \times 10^{-27} \text{ kg}$.

Answer: $2.65 \times 10^{-14} \text{ m}$.

2. Determine the velocity and kinetic energy of a neutron having de-Broglie wavelength $1 \text{ \AA}$. Given mass of neutron is $1.67 \times 10^{-27} \text{ kg}$ and Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$.

Answer: Velocity = 3970 m/s or $3.97 \times 10^3 \text{ m/s}$. Kinetic Energy = $1.32 \times 10^{-20} \text{ Joule}$.

3. Calculate the de-Broglie wavelength of an ' α ' particle accelerated through a potential difference of 200 volt.

Answer: $7.17 \times 10^{-13} \text{ m}$ or $0.0071 \text{ \AA}$.

4. Calculate the kinetic energy of an electron if its de-Broglie wavelength equals the wavelength of sodium light. Given the wavelength of sodium light $\lambda = 5893 \text{ \AA}$.

Answer: Kinetic Energy = $6.95 \times 10^{-25} \text{ Joule}$

5. An electron is bound in a one-dimensional potential box which has width $2.5 \times 10^{-10} \text{ m}$. Assuming the height of the box to be infinite, calculate the lowest two permitted energy values of the electron. ($h = 6.63 \times 10^{-34} \text{ Js}$, mass of electron $m_e = 9.1 \times 10^{-31} \text{ kg}$)

Answer: $E_1 = 6.04 \text{ eV}$ and $E_2 = 24.16 \text{ eV}$.

6. Calculate the energy difference between the first excited state and the second excited state for an electron in a one-dimensional rigid box of length 10^{-8} cm .

Answer: $E_3 - E_2 = 190 \text{ eV}$.

7. A particle confined in a one-dimensional box of width $2 \times 10^{-10} \text{ m}$ and having ground state energy $E_1 = 1.47 \times 10^{-18} \text{ Joule}$. Find the mass of the particle.

Answer: $9.32 \times 10^{-31} \text{ kg}$.

8. Find the ground state energy of an electron moving in an infinitely high one-dimensional potential box of width $1 \text{ \AA}$.

Answer: $6.02 \times 10^{-18} \text{ Joule}$ or 37.6 eV .

9. Calculate the de-Broglie wavelength of an electron whose kinetic energy is 50 eV . ($h = 6.62 \times 10^{-34} \text{ Js}$, $m = 9.1 \times 10^{-31} \text{ kg}$. $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)

Answer: $\lambda = 1.73\text{\AA}$

10. Calculate the energy difference between the ground state and the first excited state for an electron in a 1D rigid box of length 10^{-10} m.

Answer: Energy difference $= 1.81 \times 10^{-17}$ *Joule* or 112.8 *eV*.

UNIT 4

1. A Ruby Laser emits light of wavelength 694.3 nm. Calculate the frequency of emitted radiation and energy of one photon. (Planck's constant, $h = 6.626 \times 10^{-34}$ J-s, Speed of light, $c = 3 \times 10^8$ m/s)

Answer: $\nu = 4.32 \times 10^{14}$ Hz, $E = 2.86 \times 10^{-19}$ J

2. A ruby laser emits a pulse of energy 0.5 J at wavelength 694.3 nm. Find the number of photons emitted.

Answer: $N = 1.75 \times 10^{18}$ photons

3. Find the ratio of population N_2/N_1 at temperature $T = 300$ K if energy difference $E_2 - E_1 = 1$ eV. (Given: $k = 8.617 \times 10^{-5}$ eV/K).

Answer: $N_2/N_1 = 1.5 \times 10^{-17}$

4. If $N_2 = 5 \times 10^{18}$ and $N_1 = 2 \times 10^{18}$, check whether inversion exists.

Answer: Yes, population inversion exist

5. At $T = 500$ K, if $N_2/N_1 = 0.01$, find energy gap E .

Answer: $E = 0.198$ eV

6. Refractive indices of core $n_1 = 1.48$, cladding $n_2 = 1.46$. Find Numerical Aperture (NA).

Answer: $NA = 0.24$

7. Find the critical angle for core $n_1 = 1.5$, cladding $n_2 = 1.48$.

Answer: Critical angle = 81°

8. An optical fiber has core refractive index $n_1 = 1.50$ and cladding refractive index $n_2 = 1.47$. Find:

- Numerical Aperture (NA)
- Acceptance angle in air
- Maximum angle inside the fiber (refraction angle)

Answer: a. 0.298° , b. 17.3° , c. 11.5°

9. An optical fiber has: Core refractive index, $n_1 = 1.50$; Cladding refractive index, $n_2 = 1.48$. A light ray inside the core strikes the core-cladding interface at an angle of 78° (with the normal). Will the ray undergo total internal reflection?

Answer: Total internal reflection does not occur.

10. An engineer wants an optical fiber with an acceptance angle of 30° in air. The core refractive index is 1.50. Find the required cladding refractive index.

Answer: cladding refractive index = 1.414